MILITARY SPECIFICATION

FOR

COOLER, CRYOGENIC DETECTOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Cryogenic Detector Cooler critical item (Drawing 639AS3200), referred to herein as the cooler.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1420

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking for U.S. Military Property.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-704	Aircraft Electric Power Characteristics.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

639AS3200	Cooler, Cryogenic Detector.
SA492551	SW-1C Vibration Test Fixture, MARK 18.
SA2875405	Vibration Test Fixture Cooler, Cryogenic Detector.

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PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-1322	Reliability Program Provisions for Production of Sidewinder AIM-9.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S1.4-71	Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10013.)

3. REQUIREMENTS

3.1 Item description. The cooler is used to cool the infrared detector of the Sidewinder missile. The cooler operates on the split Stirling thermodynamic cycle, and consists of a compressor assembly and a remote expander assembly. A small diameter pneumatic tube interconnects the two assemblies and provides the means through which the compressor energizes the expander (see Figure 1). The cooler fits inside the Sidewinder guidance control section (GCS) and is powered from the missile launcher through the missile umbilical.

3.2 Characteristics.

3.2.1 Performance. Each cooler shall be subjected to burn-in and screen vibration and shall subsequently be subjected to and meet the performance requirements specified herein.

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3.2.1.1 Cooldown time. The time interval between the application of electrical power to the cooler and the moment the cold end temperature sensor reaches 100 degrees Kelvin (°K) shall be not greater than 3 minutes.

3.2.1.2 Steady-state operation. Steady-state operation is the powered operation following cooldown. The cooler shall meet the following steady-state operating conditions.

3.2.1.2.1 Net refrigeration capacity. Net refrigeration capacity is the rate of heat removal from external sources. The cooler shall have a 0.75 watt (W) net refrigeration capacity when the cold end temperature sensor is at 75 ±5°K when operating from -40 to +70 degrees Celsius (°C).

3.2.1.2.2 Temperature control. The cooler shall maintain the cold end temperature sensor below 75°K.

3.2.1.3 Electrical operating requirements.

3.2.1.3.1 Input voltage. The cooler shall meet all performance requirements using 115 volt (V), 400 hertz (Hz), single phase electrical power conforming to MIL-STD-704, Table I, Category B.

3.2.1.3.2 Start-up and steady-state power consumption. When operating with a power input of 108 ±0.5 V the following conditions shall be met:

a. The current shall not exceed 3 amperes (A) during the first 3 seconds of cooler operation.

b. The current shall not exceed 2 A from 3 seconds until 1 minute after power application to the cooler.

3.2.1.3.3 Steady-state power consumption. The operating current shall not exceed 1 A after 1 minute of operation.

3.2.2 Physical characteristics.

3.2.2.1 Weight. The cooler shall weigh between 5.5 and 13.5 pounds.

3.2.2.2 Structural strength.

3.2.2.2.1 Proof pressure. The cooler shall withstand, without degradation of performance or permanent structural deformation, an applied internal pressure 2,100 ±30 pounds per square inch gage (psig).

3.2.2.2.2 Burst pressure. The cooler shall not rupture with an applied internal pressure equal to 2.5 times peak pressure. The cooler is not required to operate subsequent to burst pressure test.

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3.2.2.3 Leak rate. Leakage of the cryogen from the system with the cooler charged in accordance with Drawing 639AS3200, shall not exceed 2×10^{-8} standard cubic centimeters per second.

3.2.3 Reliability. A reliability assurance program shall be planned, implemented, conducted, and reported in accordance with AD-1322 (see 6.2.1).

3.2.3.1 Reliability (mean-time-between-failures (MTBF)) demonstration. Unless otherwise specified in the contract or purchase order (see 6.2.1), coolers supplied in accordance with this specification shall be capable of demonstrating a specified MTBF (θ_o) of 1,300 hours when tested in accordance with MIL-STD-781, Test Plan XIVC for first article testing and Test Plan IVC for production lot testing.

3.2.3.2 MTBF performance requirements. The MTBF performance requirements shall be as specified in Table I when subjected to the test profile of Figure 2.

TABLE I. Performance test sequence.

Test	Requirement paragraph	Method paragraph
Cooldown	3.2.1.1	4.4.2.1
Net refrigeration capacity	3.2.1.2.1	4.4.2.2
Temperature control	3.2.1.2.2	4.4.2.3
Input voltage	3.2.1.3.1	4.4.2.4
Start-up	3.2.1.3.2	4.4.2.5
Power consumption	3.2.1.3.3	4.4.2.6
Leak rate	3.2.2.3	4.4.2.9

3.2.3.3 MTBF determination (first article and production lot). The cooler operating time shall be used in MTBF determination. The operating time for each cooler subjected to reliability testing shall be not less than one-half the average operating time of all coolers which completed the reliability (MTBF determination) testing. For the purposes of MTBF demonstration, the cooler operating time is that portion of the test profile of Figure 2 in which power is supplied to the cooler. No cooler shall receive maintenance of any kind during reliability testing.

3.2.3.4 Operating life. The coolers supplied in accordance with this specification shall be capable of demonstrating a minimum operating life of 500 hours. Operating life is the cumulative cooler operating hours during which the cooler meets the performance requirements specified herein, no relevant failures are experienced, and no maintenance is performed.

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3.2.3.5 Operating life performance requirements. The operating life performance requirements shall be as specified in Table I when subjected to the test profile of Figure 2.

3.2.3.6 Operating life determination. The cooler operating time shall be used in operating life determination. Operating life shall be in accordance with 3.2.3.4. Each cooler selected for life testing shall receive no less than 500 hours on-time. No maintenance shall be performed on any cooler undergoing life testing.

3.2.4 Environmental conditions. The cooler shall meet all performance requirements of 3.2.1 during or subsequent to exposure to any single environmental condition or combination of environmental conditions specified in Table II, Group II or Table III, Group II.

TABLE II. First article tests and inspections.

Test or inspection	Requirements paragraph	Methods paragraph
Group I: Sixteen coolers		
Construction	3.3	
Visual and mechanical		4.4.1
Cooldown time	3.2.1.1	4.4.2.1
Net refrigeration capacity	3.2.1.2.1	4.4.2.2
Temperature control	3.2.1.2.2	4.4.2.3
Input voltage	3.2.1.3.1	4.4.2.4
Start-up	3.2.1.3.2	4.4.2.5
Power consumption	3.2.1.3.3	4.4.2.6
Proof pressure	3.2.2.2.1	4.4.2.7
Leak rate	3.2.2.3	4.4.2.9
Group II: Six coolers		
Environmental		
Low temperature (operating)	3.2.4	4.4.4.1
Shock	3.2.4	4.4.4.2
High temperature (operating)	3.2.4	4.4.4.3.1
Temperature environment	3.2.4	4.4.4.3.2
Temperature shock	3.2.4	4.4.4.3.3
Vibration	3.2.4	4.4.4.4
Group III: Ten coolers		
Reliability	3.2.3.1, 3.2.3.2, 3.2.3.3	4.2.4, 4.4.3.1
Burst pressure (one cooler only)	3.2.2.2.2	4.4.2.8

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TABLE III. Production tests and inspections.

Test or inspection	Requirements paragraph	Methods paragraph
Group I: All coolers		
Construction	3.3	
Visual and mechanical		4.4.1
Cooldown time	3.2.1.1	4.4.2.1
Net refrigeration capacity	3.2.1.2.1	4.4.2.2
Temperature control	3.2.1.2.2	4.4.2.3
Input voltage	3.2.1.3.1	4.4.2.4
Start-up	3.2.1.3.2	4.4.2.5
Power consumption	3.2.1.3.3	4.4.2.6
Proof pressure	3.2.2.2.1	4.4.2.7
Leak rate	3.2.2.3	4.4.2.9
Group II: QCT lot sample		
Environmental		
Low temperature (operating)	3.2.4	4.4.4.1
High temperature (operating)	3.2.4	4.4.4.3.1
Temperature environment	3.2.4	4.4.4.3.2
Group III: Twelve coolers/lot		
Reliability	3.2.3.1, 3.2.3.2, 3.2.3.3	4.2.4, 4.4.3.2
Burst pressure (one cooler only)	3.2.2.2.2	4.4.2.8

3.3 Construction. The cooler shall be fabricated and assembled in accordance with Drawing 639AS3200 and associated documents.

3.3.1 Standards of manufacture.

3.3.1.1 Special working conditions. As a minimum, fabrication and assembly operations shall be in accordance with Figure 3.

3.3.2 Identification and product marking. The cooler shall be marked in accordance with MIL-STD-130. A serial number shall be stenciled on each cooler produced.

3.4 First article. When specified, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2.3).

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3.5 Workmanship. The cooler, including all parts and accessories, shall be constructed and finished in such a manner as to ensure compliance with all requirements of this specification. Prior to and after final assembly, all parts and components shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of tests and inspections. Tests and inspections specified herein are classified as follows:

a. First article tests and inspections (see 4.2.1).

b. Individual tests and inspections (see 4.2.2).

c. Quality conformance tests (QCT) (see 4.2.3).

d. Reliability and life tests (see 4.2.4).

4.1.2 Sampling. Sampling for each of the tests and inspections of 4.1.1 shall be as follows:

a. First article tests and inspections. The sample shall consist of 16 coolers as specified in 4.2.1. Following the Group I tests of Table II, the sample shall be subdivided into two groups. One group of six shall be subjected to the Group II tests of Table II and the remaining group of 10 shall be subjected to the Group III tests of Table II.

b. Individual tests and inspections. Each cooler shall be subjected to the tests and inspections specified in 4.2.2.

c. Quality conformance tests. Each cooler of each production lot shall meet the requirements of 4.2.2 prior to forming the QCT sample. Random samples of coolers shall be selected from each

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production lot as follows and then subjected to the Group II tests of Table III:

(1) One to 49 coolers: two samples.

(2) 50 to 99 coolers: three samples.

(3) 100 to 199 coolers: four samples.

(4) 200 to 250 coolers: six samples.

d. Reliability tests. The reliability tests shall be as specified in Table II, Group III for first article testing and Table III, Group III for production lot sample testing. Sampling for reliability shall be as follows:

(1) During the first article testing: Ten of the coolers selected as the first article sample (see 4.1.2a and 4.2.1) shall be subjected to the reliability test of Table II, Group III.

(2) During production lot testing: In addition to the QCT sample of 4.1.2c, 12 coolers which have met the requirements of 4.2.2 shall be selected from the production lot and then subjected to the Group III tests of Table III.

e. Life tests. When specified in the contract or purchase order (see 6.2.1), three coolers shall be identified as life test units from those selected in 4.1.2.d(2) prior to the commencement of reliability testing. After completion of the reliability tests (see 4.2.4.2), the life test units shall be subjected to life testing in accordance with 4.4.3.3. Operating time accumulated during reliability testing will count towards achievement of the life test. A life test report (see 6.2.2) shall be prepared for the procuring activity.

4.1.2.1 Production lot. A production lot shall consist of a group of coolers produced by the same process and under the same conditions. Lots are only formed from those units that have passed the requirements of the Group I, Table III tests and inspections (see 4.2.2).

4.1.2.2 Lot storage. After the test sample quantity of coolers of a lot have been selected and distributed in accordance with 4.1.2, the remainder of the lot shall be stored in bonded storage pending acceptance or rejection.

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4.1.3 Test sample disposition. Upon completion of any of the tests and inspections specified herein, and after notification of test results approval for that test by the procuring activity, the test samples shall be dispositioned as follows:

a. First article sample. The six coolers subjected to the Group II, Table II tests shall be marked for identification as the first article environmental sample and shall be delivered to the cognizant design activity. The 10 coolers subjected to the Group III, Table II tests shall be marked for identification as the first article reliability demonstration sample and shall be delivered to the cognizant design activity.

b. Production lot samples. The coolers selected as the production lot sample (Group II and III, Table III tests) shall be refurbished as appropriate and tested in accordance with a refurbishment plan (see 6.2.2) prepared by the contractor for the review and approval of the procuring activity. Coolers used for the burst pressure test shall be delivered to the cognizant design activity without refurbishment. Refurbished units shall be returned to the production lot.

c. Life test samples. The three coolers selected from each production lot for life testing shall be marked for identification as production lot life test samples along with the production lot number from which they were selected, and shall be delivered to the cognizant design activity.

4.1.4 Failure reporting, analysis, and corrective action. The contractor shall perform failure reporting, failure analysis, and corrective action in accordance with AD-1322 and as specified in the contract or purchase order (see 6.2.2). Corrective action for a design, parts, or materials related failure shall require approval by the procuring activity. Cooler operation time shall be monitored and recorded at time of failure.

4.1.5 Relevant failures. A relevant failure is any failure except those classified and listed herein as a nonrelevant failure. Concurrence by the cognizant design activity shall be required for classification of all failures experienced during first article reliability and production lot life testing.

4.1.6 Nonrelevant failure.

a. Any failure caused by test equipment.

b. Any failure that results from operational or procedural error in testing.

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c. Secondary failures when they are confirmed to be induced by failure of another part. However, at least one primary (relevant) failure shall be charged when a secondary failure is claimed.

d. When a failure results from a design deficiency which has been determined to exist, and concurred with in writing by the procuring activity, provided that the required corrective action for the design deficiency does not apply to the lot in question.

e. Repeated failures for which corrective action has been documented but, by direction of the procuring activity in writing, the corrective action is not incorporated into the cooler lot where the repeated failures occur.

f. Any failure resulting from rework, repair, or investigation of failures during environmental or MTBF testing if detected prior to resubmission to environmental or MTBF testing.

4.2 Tests and inspections.

4.2.1 First article. Unless otherwise specified in the contract or purchase order, a first article sample of 16 coolers shall be delivered to the testing activity designated in the contract or purchase order for first article testing (see 6.2.3). The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample is at the risk of the contractor. After approval of the first article sample, changes in materials, parts, procedures, design, or processes shall require written approval of the procuring activity.

4.2.1.1 First article tests. The coolers selected as the first article sample (see 3.4 and 4.2.1) shall be subjected to the tests and inspections in accordance with Table II.

4.2.1.2 First article rejection. Failure of the first article sample to meet any of the requirements specified herein shall cause rejection of the first article sample.

4.2.1.3 Test plans. Unless otherwise specified in the contract or purchase order (see 6.2.1), detailed test plans (see 6.2.2) in accordance with MIL-T-18303 shall be prepared by the contractor and receive approval of the procuring activity prior to the commencement of testing.

4.2.1.4 Tests reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), test reports (see 6.2.2) in accordance with MIL-STD-831 shall be prepared by the contractor for approval of the procuring activity for all first article tests. Test reports are required for environmental and MTBF tests performed on the first article sample.

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4.2.2 Individual tests and inspections. Each cooler of each production lot shall be subjected to the Group I tests and inspections as specified in Table III. Failure of any cooler to pass any of the tests specified shall result in the rejection of the cooler and its removal from the lot.

4.2.3 Quality conformance tests. Each cooler in the production lot shall have been subjected to and shall have passed the individual tests of 4.2.2 prior to forming a lot sample for QCT. QCT shall consist of the tests specified in Group II of Table III. The order of tests is advisory. These tests shall be performed on a sample of each production lot in accordance with the sampling requirements of 4.1.2c.

4.2.3.1 Production lot rejection. Failure of any cooler in the lot sample to meet any of the requirements specified herein shall reject the lot represented.

4.2.3.2 Production QCT test reports. For each production lot offered for acceptance, the contractor shall prepare a test report (see 6.2.2) for the selected samples that are subjected to the environmental and life tests.

4.2.4 Reliability and life tests.

4.2.4.1 Reliability tests during first article testing. Each cooler selected for reliability (MTBF) tests shall be subjected to and shall pass the tests of Table II, Group I prior to commencement of any reliability testing. The coolers selected for first article reliability testing (see 4.1.2d(1)) shall be subjected to and shall pass the tests of Table II, Group III in the order shown.

4.2.4.2 Reliability tests during production lot testing. The coolers selected for production lot reliability testing (see 4.1.2d(2)) shall be subjected to and shall pass the tests of Table III, Group III in the order shown. Each cooler selected for reliability testing shall be subjected to and shall pass the individual tests of 4.2.2 prior to commencement of reliability testing.

4.2.4.3 Reliability test-rejection.

a. Failure of the first article reliability test sample to meet any of the requirements specified herein shall reject the first article sample (see 4.4.3.1).

b. Failure of the production lot reliability test sample to meet any of the requirements specified herein shall reject the lot represented (see 4.4.3.2).

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4.3 Test conditions.

4.3.1 Calibration of test equipment. All electrical, electronic, and mechanical measuring equipment used in acceptance testing and mechanical assembly shall be calibrated in accordance with a system meeting the requirements of MIL-STD-45662, except in the "Calibration procedures" paragraph, the sentence "As a minimum, the procedures shall specify the accuracy of the instrument being calibrated and the accuracy of the standards used," and the contents of the "Evaluation of suspect product" paragraph in its entirety, shall not be applicable, by using secondary laboratory standards for determination of accuracy. The secondary laboratory standards shall have been calibrated against primary standards with recalibration on a regularly scheduled basis. The referred source for the primary standards shall be certified by the National Bureau of Standards. The contractor shall certify that all test equipment has been so calibrated before testing begins. Certification to this effect (see 6.2.2) shall accompany all reports of first article, quality conformance testing, and reliability and life testing (see 4.2.1.4 and 4.2.3.2). Recalibration shall be made at necessary intervals or as required by the Government representative.

4.3.1.1 Test station certification. Unless otherwise specified in the contract or purchase order (see 6.2.1), all test stations used for measuring final acceptance requirements of the cooler shall require certification by the procuring activity.

4.3.1.2 Special test equipment. Unless specific exceptions are made by the procuring activity or unless values are given for individual tests, all special test equipment employed as a means for testing the cooler shall be of such a nature and design that any errors produced do not exceed 10 percent of the maximum allowable tolerance established for the value to be measured.

4.3.2 Test facility conditions. Unless otherwise specified herein, the cooler shall be subjected to the tests specified herein under the following conditions:

a. Temperature. 18 to 29°C.

b. Altitude. Ground elevation of test facility.

c. Vibration. Condition at test facility.

d. Humidity. Ambient (not exceeding 95 percent relative).

4.3.3 Test equipment characteristics. Unless otherwise specified herein, the test equipment used shall have the following characteristics:

a. Temperature sensors shall be calibrated to an accuracy of ±2° over the range of 40 to 115°K.

b. The dummy GCS shall conform to Drawing SA2875405.

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4.4 Test methods. Prior to being subjected to the individual tests specified herein, each cooler shall complete a burn-in cycle in accordance with 4.4.5 and a screening vibration test in accordance with 4.4.6. Any alternate test method shall have prior approval of the Government.

4.4.1 Visual and mechanical examinations. The cooler shall be examined for conformance with the requirements of Drawing 639AS3200 and associated documents, and 3.5.

4.4.2 Performance tests. Unless otherwise specified herein, the cooler shall be held with the longitudinal axis in a horizontal position and mounted in a holding fixture subject to approval by the procuring activity. Cooldown time (see 4.4.2.1) and start-up (see 4.4.2.5) tests may be conducted concurrently. When the cooldown time and start-up tests are performed together, the input voltage shall be 108 ±0.5 V.

4.4.2.1 Cooldown time. The cooler shall be nonoperating and stabilized in an environment of 18 to 29°C for a period of not less than 30 minutes. The Dewar temperature sensor output shall be recorded on a strip chart recorder commencing 10 ±5 seconds prior to compressor power turn-on and continued until the Dewar temperature sensor reaches 75°K. A 0.25 ±0.01 W thermal load shall be applied to the expander cold end at the same time power is applied to the compressor. To be acceptable, the cooler shall meet the requirements of 3.2.1.1.

4.4.2.2 Net refrigeration capacity. The cooler shall be stabilized in the environment shown prior to making the refrigeration capacity measurement. The heat load shall be 0.75 ±0.01 W. The cooler shall be operated and allowed to stabilize before the temperature and heater power readings are recorded. This test shall be performed at the following three temperatures:

a. Ambient

b. -40 ±5°C

c. +70 ±5°C

To be acceptable, the cooler shall meet the requirements of 3.2.1.2.1.

4.4.2.3 Temperature control. After cooldown has been achieved, the cooler shall be operated for 60 ±5 minutes in an environment of 18 to 29°C with the ambient temperature and compressor input voltage held constant. A 0.125 ±0.010 W load shall be applied to the expander cold end. The Dewar temperature sensor output shall be recorded on a strip chart recorder during the 60-minute period. To be acceptable, the cooler shall meet the requirements of 3.2.1.2.2.

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4.4.2.4 Input voltage. The cooler shall be connected to a 400 Hz, single-phase, variable voltage supply. The tests of 4.4.2.1, 4.4.2.3, and 4.4.2.6 shall be performed with an input voltage level of 108.0 ±0.5 volts root-mean-square.

4.4.2.5 Start-up. The cooler shall be connected to the voltage supply of 4.4.2.4. Instrumentation shall monitor motor current drain from 30 ±5 seconds prior to system power-on until 1 minute after system power-on. To be acceptable, the cooler shall meet the requirements of 3.2.1.3.2.

4.4.2.6 Power consumption. The cooler shall be connected to the voltage supply of 4.4.2.4 and the voltage adjusted to 115.0 ±0.5 V. The current shall be measured during start-up and during 10 minutes of steady-state operation. To be acceptable, the cooler shall meet the requirements of 3.2.1.3.2 and 3.2.1.3.3.

4.4.2.7 Proof pressure. The system shall be charged to 2,100 ±30 psig for 30 ±3 minutes with helium gas. To be acceptable, the cooler shall meet the requirements of 3.2.2.2.1. At the end of 30 minutes, the charge pressure shall be reduced to the system charge pressure and subjected to the conditions of 4.4.2.1 and 4.4.2.3. This test need only be performed once.

4.4.2.8 Burst pressure. The cooler, contained in a protective enclosure, shall be pressurized to 2.5 times peak pressure for 10 ±1 minutes using an appropriate medium. To be acceptable, the cooler shall meet the requirements of 3.2.2.3.2.

4.4.2.9 Leak rate. The cooler shall be subjected to the gas leakage test specified in MIL-STD-202, Method 112 B, Test Condition C, Procedure IV. The cooler may be removed from the holding fixture and may be oriented in any position for this test. To be acceptable, the cooler shall meet the requirements of 3.2.2.3.

4.4.3 Reliability and life tests.

4.4.3.1 First article reliability demonstration test. The first article reliability demonstration test shall be conducted in accordance with MIL-STD-781, Test Plan XIVC (see 3.2.3.1) and the following:

a. A test sample of 10 coolers shall be selected in accordance with 4.1.2d(1).

b. Environmental exposure and test profile shall be in accordance with the test cycle shown in Figure 2 while mounted into the vibration test fixture (Drawing SA2875405).

c. The performance tests of Table I shall be performed at the conclusion of every 6 ±3 test cycles shown in Figure 2. The cooler shall conform to the performance requirements of Table I.

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d. Six hours of cooler operating time shall be accumulated during each 8-hour cycle shown in Figure 2. In the event a failure is detected during chamber testing, all operating time accrued to the point of failure shall be counted. In the event a failure is detected during performance verification, the failure shall be assumed to have occurred halfway between the point of detection and the last successful performance verification. All operating time and all relevant failures of all coolers committed to MTBF testing shall be used in MTBF determination.

e. Failure reporting and analysis (see 6.2.2) shall be required with failure categories given in 4.1.5.

4.4.3.2 Production lot reliability demonstration test. The production lot reliability demonstration test shall be conducted in accordance with MIL-STD-781, Test Plan IVC (see 3.2.3.1) and the following:

a. A test sample of 12 coolers shall be selected in accordance with 4.1.2d(2).

b. Environmental exposure and test profile shall be in accordance with the test cycle shown in Figure 2 while mounted into the vibration test fixture (Drawing SA2875405).

c. The performance tests of Table I shall be performed at the conclusion of every 6 ±3 test cycles shown in Figure 2. The cooler shall conform to the performance requirements of Table I.

d. Six hours of cooler operating time shall be accumulated during each 8-hour cycle shown in Figure 2. In the event a failure is detected during chamber testing, all operating time accrued to the point of failure shall be counted. In the event a failure is detected during performance verification, the failure shall be assumed to have occurred halfway between the point of detection and the last successful performance verification. All operating time and all relevant failures of all coolers committed to MTBF testing shall be used in MTBF determination.

e. Failure reporting and analysis (see 6.2.2) are required for production lot reliability assessment purposes with failure categories given in 4.1.5.

4.4.3.3 Operating life test. The operating life test shall be conducted as follows:

a. A test sample of three coolers shall be selected in accordance with 4.1.2e.

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b. Environmental exposure and test profile shall be in accordance with the test cycle shown in Figure 2 while mounted into the vibration test fixture (Drawing SA2875405).

c. The performance tests of Table I shall be performed at the conclusion of every 6 ±3 test cycles shown in Figure 2. Each cooler shall conform to the performance requirements of Table I.

d. Six hours of cooler operating time shall be accumulated during each 8-hour cycle shown in Figure 2. In the event a failure is detected during chamber testing, all operating time accrued to the point of failure shall be counted. In the event a failure is detected during performance verification, the failure shall be assumed to have occurred halfway between the point of detection and the last successful performance verification.

e. Testing shall continue until wear out of each cooler occurs or 1,000 hours on each cooler is reached. During operating hours 501 through 1,000, a failure of any kind shall not constitute grounds for rejection; however, failure reporting and analysis (see 6.2.2) are required for production lot reliability assessment purposes with failure categories given in 4.1.5.

f. Failure of the operating life test shall result if any of the following conditions occur:

(1) Any cooler failure due to wear out prior to 500 operating hours.

(2) Occurrence of two or more failures (other than wear out) during the operating life test.

4.4.4 Environmental. Unless otherwise specified in the contract or purchase order (see 6.2.1) the Group II test of Tables II or III shall be performed on first article and production lot samples, respectively. Unless otherwise specified herein, the cooler shall be mounted in the vibration test fixture (Drawing SA2875405) for each environmental test. Immediately prior to performing the environmental tests specified herein, the cooler shall be subjected to the tests specified in Group I, Tables II or III to establish a record of initial performance under standard conditions. The tests of Table I shall be performed at the conclusion of each environmental test for the purpose of determining acceptability and for comparing effects of the severity of the environment. The performance data obtained at the completion of each environmental test shall serve as initial record data for the next test, providing the results are acceptable and within tolerance. When the test procedure requires the test hardware to be system operating, the cooler shall be

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powered and the Dewar temperature sensor continually monitored. Unless otherwise specified herein, for the cooler to be acceptable, the Dewar temperature sensor shall be maintained below 75°K during each system operating period and at the conclusion of each test procedure, the requirements of Table I shall be met.

4.4.4.1 Low temperature (operating). The cooler shall be placed in a test chamber suitable for performing the tests of Table I. The internal temperature of the chamber shall be reduced to -40 ±5°C, and the cooler shall remain at this temperature for a minimum of 4 hours. The cooler shall be operated in the chamber at -40 ±5°C and tested in accordance with Table I. At the conclusion of the environmental test, the cooler shall be subjected to and shall pass the tests of Table I.

4.4.4.2 Shock. The cooler shall be mounted in a fixture capable of directing shock pulses along each of the three mutually perpendicular axes of the cooler. All transverse shock tests shall have the acceleration transducer mounted on the cooler fixture within 1 inch of the combined cooler and fixture center of gravity. When the cooler is subjected to the shock test along its longitudinal axis, the acceleration transducer shall be mounted in the base of the shock table. The cooler shall be nonoperating during the shock test.

4.4.4.2.1 Procedure for shock test. The cooler shall be subjected to two shock pulses, 47.5 ±2.5 gravity units peak amplitude, 21 ±3 milliseconds duration (see Figure 4) half-sine pulse in each of the six directions along the three mutually perpendicular axes for a total of 12 shock pulses. At the conclusion of shock tests, the cooler shall be subjected to and shall pass the tests of Table I.

4.4.4.3 Temperature tests.

4.4.4.3.1 High temperature (operating). The cooler shall be placed in a test chamber suitable for performing the tests of Table I. Internal temperature of the test chamber shall be raised to +74 ±5°C. The cooler shall be maintained at this temperature for a minimum of 4 hours. At the end of the 4-hour period, the cooler shall be operated in the chamber at +74 ±5°C and tested in accordance with Table I. At the conclusion of the environmental test, the cooler shall be subjected to and shall pass the tests of Table I.

4.4.4.3.2 Temperature environment. The cooler shall be subjected to the heating program in accordance with 4.4.4.3.2.1. The cooler shall meet the requirements of 3.2.1.2.2 during each test cycle.

4.4.4.3.2.1 Test procedure. The cooler shall be placed in a heating facility, as approved by the cognizant Government activity, designed to apply the required temperatures to all external surfaces of

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the cooler. The control thermocouple shall be mounted on the external surface at a point 6 inches aft of the rear edge of the dome locking ring. The programmed heating shall be as follows:

a. Stabilize cooler for 2 hours at +54 ±5°C.

b. Increase temperature linearly to +100 ±5°C in 5 minutes ±30 seconds.

c. Hold at +100 ±5°C for 15 minutes ±30 seconds.

d. Increase temperature to +140 ±5°C within a maximum of 2 minutes.

e. Hold at +140 ±5°C for 5 minutes ±30 seconds.

f. Allow coolers to cool to room temperature.

This constitutes one cycle of approximately 2.5 hours. The number of complete cycles shall be five. At the conclusion of the fifth cycle and while stabilized at room temperature, the cooler shall be subjected to and shall pass the tests of Table I.

4.4.4.3.3 Temperature shock. The cooler shall be subjected to the temperature shock tests, conducted in accordance with the following modification of MIL-STD-810, Method 503. The coolers shall be non-operating during the test.

Step 1. The cooler shall be placed in a suitable high temperature chamber whose internal temperature is +74 ±5°C and maintained at this temperature for not less than 2 hours.

Step 2. The cooler shall be transferred within 5 minutes to a cold chamber whose internal temperature is -54 ±5°C and maintained at this temperature for not less than 2 hours.

Step 3. The cooler shall be returned to the high temperature chamber within 5 minutes after removal from the low temperature chamber.

Steps 1 through 3 shall constitute one cycle of temperature shock testing. The cooler shall be subjected to five cycles of testing and, at the conclusion of the fifth 2-hour cooling period of Step 2, the cooler shall be removed from the test chamber and allowed to return to room temperature. The cooler shall be subjected to and shall pass the tests of Table I.

4.4.4.4 Vibration. The vibration tests shall consist of ambient temperature vibration (18 to 29°C) and low temperature vibration (-35 to -45°C). Provisions for conducting these tests shall include instrumentation that is calibrated at the temperature extremes required for the test.

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4.4.4.4.1 Mounting for vibration. The cooler, while assembled in an instrumented GCS, shall be mounted in the fixture in accordance with Drawing SA492551. The fixture shall be attached to a vibration table for vibration in each of the three mutually perpendicular axes of the instrumented GCS. Vibration excitation along each axis (see Figure 5) shall be applied to the table with the geometric center of the fixture parallel with the vibration exciter armature centerline.

a. Verify that magnetic coupling between the exciter and cooler does not affect performance of the cooler during vibration. This may be accomplished by decoupling the cooler from the exciter, maintaining the same approximate location of the cooler in respect to the exciter head, and sweeping the exciter through the specified frequency range and power level for 4 minutes each in the three mutually perpendicular planes. The temperature of the Dewar shall not change by more than 3°K.

b. The applied units of gravity squared per hertz level shall be within ±3 decibels of the specified input level.

c. Prior to any testing of the cooler, cross motion acceleration measurements shall be made of the fixture. Such measurements shall be made utilizing the complete vibration system assigned for the purpose of testing the cooler. Cross motion acceleration shall be conducted with a dummy cooler, installed in the fixture in accordance with Drawing SA492551. Cross motion measurements shall be made in the longitudinal and transverse axes with cross motion response measured at the accelerometer locations specified in Drawing SA492551. Cross motion acceleration shall be measured with an input to the fixture in accordance with Figure 5. Cross motion acceleration shall be not greater than 50 percent of the input in a nonresonant mode (1:2), not greater than 3 times the input in a resonant mode. A permanent record shall be made of cross motion acceleration as well as a plot of the control input power level. These records shall be made available upon request from the cognizant Government activity.

Conformance to a, b, and c shall be approved by the cognizant Government representative.

4.4.4.4.2 Procedure for vibration. The instrumented GCS, mounted in accordance with 4.4.4.4.1, shall be subjected to random vibration as specified in Figure 5 for a period of 10 minutes per axis, three axes at each of the following temperatures: ambient and -40 ±5°C. To be acceptable, the cooler shall meet the requirements of 3.2.1.2.2 during vibration. At the conclusion of vibration testing, the requirements of Table I shall be met.

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4.4.5 Burn-in. Prior to the performance tests of 4.4.2, each cooler shall go through two burn-in cycles as shown in Figure 6. To be acceptable, the cooler shall meet the requirements of 3.2.1.1 and 3.2.1.2.2 during the periods the cooler is on.

4.4.6 Screening vibration. Prior to the performance tests of 4.4.2, each cooler shall go through a screening vibration test. During vibration the cooler shall be mounted in a vibration fixture approved by the procuring activity.

4.4.6.1 Mounting for vibration. The fixture shall be attached to a vibration table for vibration in each of the three mutually perpendicular axes. Vibration excitation along each axis shall be applied to the table with the geometric center of the fixture parallel with the vibration exciter armature centerline.

4.4.6.2 Procedures for vibration screen. The cooler mounted in accordance with 4.4.6.1 shall be vibrated along each of the three mutually perpendicular axes. The random vibration shall be for 5 minutes in each axis with inputs as shown in Figure 5.

4.5 Preparation for delivery. Preservation, packaging, packing, and marking of the cooler shall be as specified in Section 5.

5. PACKAGING

5.1 Preservation and packaging. Levels of preservation and packaging shall be in accordance with FED-STD-102.

5.1.1 Level C. Containers shall comply with Consolidated Freight Rules or other common carrier regulations applicable to the mode of transportation. The preservation and packaging shall include cleaning, drying, and preservation methods adequate to prevent deterioration; appropriate protective wrappings; package cushioning; and interior containers (excluding shipping containers).

5.2 Packing. Packing shall cover the exterior shipping container, necessary cushioning, and weatherproofing.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cooler is intended for use in infrared guidance seekers.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. A reliability assurance program (see 3.2.3).

c. Reliability MTBF demonstration, if other than specified in 3.2.3.1.

d. Responsibility for inspection, if other than specified in 4.1.

e. Life test units (see 4.1.2e).

f. Test plans, if required (see 4.2.1.3 and 6.2.2).

g. Test reports and distribution if required (see 4.2.1.4 and 6.2.2).

h. Test station certification, if required (see 4.3.1.1).

i. Environmental tests, if other than specified in 4.4.4.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirements	Applicable DID	Option
4.1.2e	Test Reports	DI-T-2072	---
4.2.1.4	Test Reports	DI-T-2072	---
4.2.3.2	Test Reports	DI-T-2072	---
4.1.3b	Refurbishment Plan	UDI-S-20526	---
4.1.4	Failure Analysis Reports	DI-R-7039	---
4.4.3.1e	Failure Analysis Reports	DI-R-7039	---
4.4.3.2e	Failure Analysis Reports	DI-R-7039	---
4.4.3.3e	Failure Analysis Reports	DI-R-7039	---
4.2.1.3	Test Plans	DI-T-5204	---
4.3.1	Certification	UDI-A-23264B	---

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.3 First article. When a first article is required, the item will be tested and should be a first article sample. The first article should consist of 16 coolers, as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.3 Definitions.

6.3.1 Standard cubic centimeters. Standard cubic centimeters equal volume of gas measured in cubic centimeters corrected to +24 ±2°C and +30.0 ±5.0 inches of mercury pressure.

6.3.2 Corrective action. Corrective action is a documented design, process, procedure, or materials change to correct the true cause of the failure. Part replacement with a like item, by itself, does not constitute appropriate corrective action.

Preparing activity:
Navy (AS)
(Project 1420-N254)

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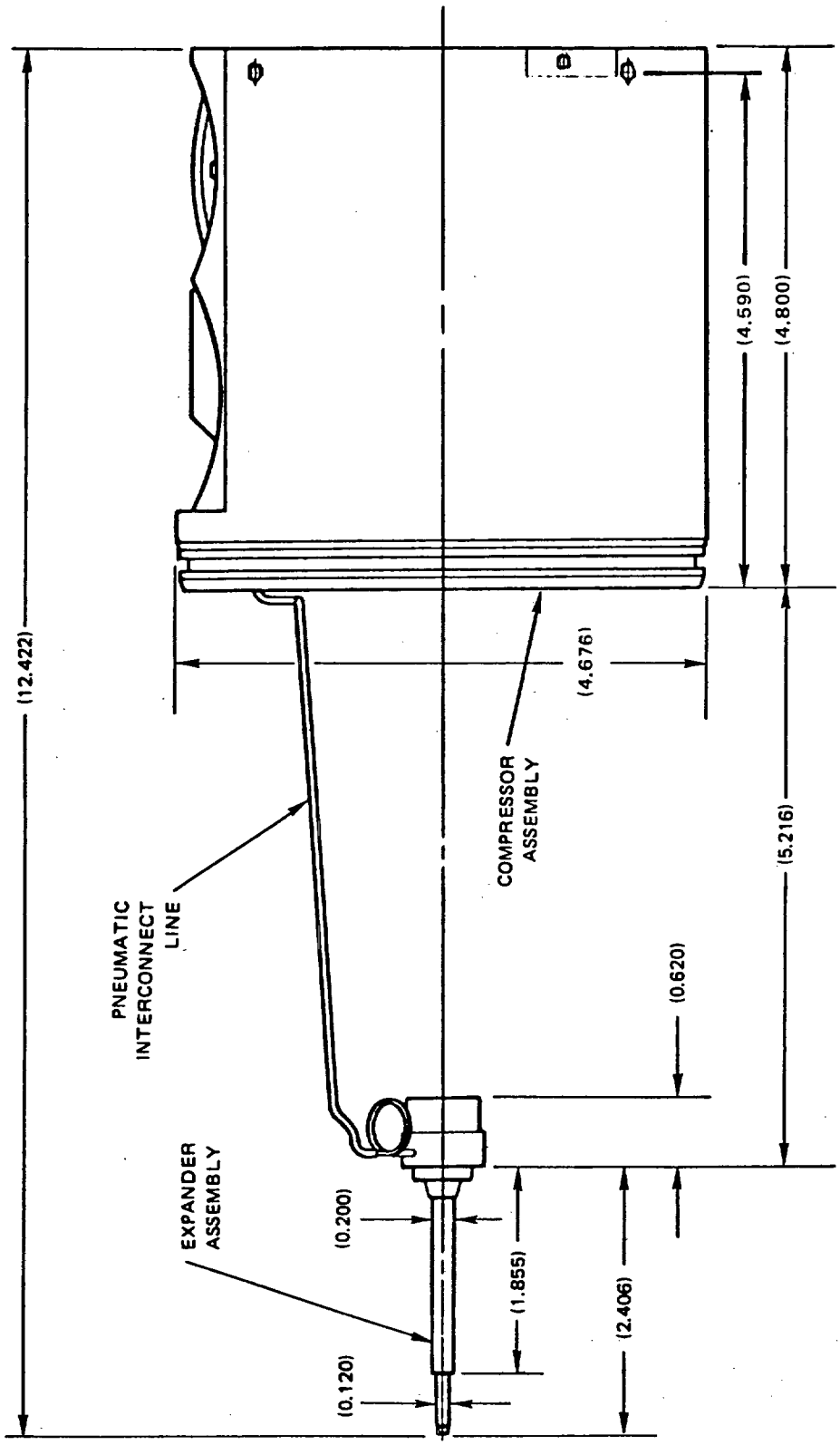
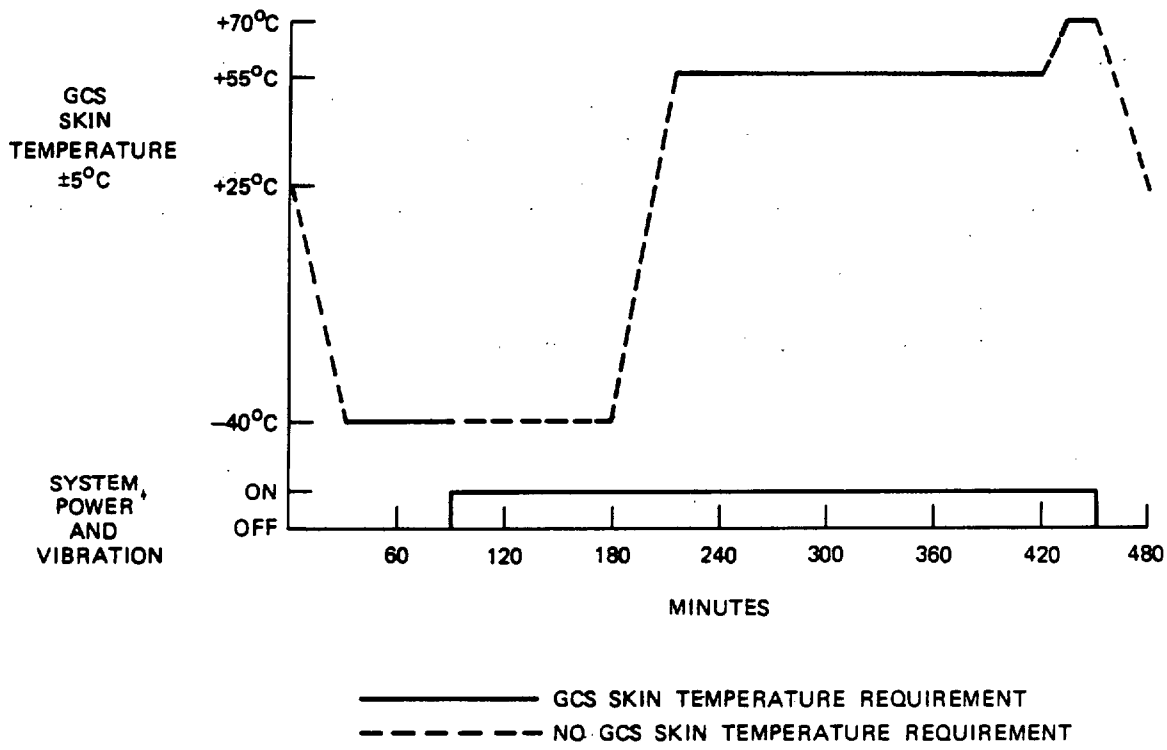


FIGURE 1. Cooler assembly.

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MTBF TEST CYCLE

STEP	SEQUENTIAL TEST TIME PERIOD (MINUTES) T	TIME DURATION (MINUTES)	TEMPERATURE ACTIVITY	TEMPERATURE REQUIREMENT ±5°C	SYSTEM POWER		VIBRATION
					ON TIME	OFF TIME	
1	0 TO 30	30	TRANSITION	+25° TO -40°		30	OFF
2	30 TO 90	60	DWELL	-40°		60	OFF
3	90 TO 180	90	MAINTAIN CHAMBER	≤ -40°	90		ON
4	180 TO 210	30	TRANSITION	TO +55°	30		ON
5	210 TO 420	210	DWELL	+55°	210		ON
6	420 TO 435	15	TRANSITION	TO +70°	15		ON
7	435 TO 450	15	DWELL	+70°	15		ON
8	450 TO 480	30	TRANSITION	TO +25°		30	OFF
TOTAL		480			360	120	

FIGURE 2. Reliability and life testing environmental profile.

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NOTES:

1. GCS skin temperature shall be measured at a point 11.60 inches forward of the aft end of the GCS housing, 180 degrees from the 0 degree mechanical reference notch.
2. A thermal survey may be conducted in accordance with MIL-STD-781, Thermal Survey, on one GCS. If this test is conducted, the GCS temperature stabilization for the upper temperature (+70°C) shall be for 15 minutes.
3. System power input voltages shall be in accordance with the external power requirements of this specification.
4. Chamber temperature during Step 3 shall be maintained below -40°C.
5. A performance test shall be made every 6 ± 3 test cycles.
6. Vibration profile A shall be applied continuously during the test cycle in which the GCS is energized. Vibration profiles B and C shall be applied as defined in the MTBF vibration application shown below. Input vibration level shall be monitored on vibration fixture. Transition from one vibration level to another vibration level shall be with minimal transitory time.

GCS MTBF VIBRATION PROFILES

PROFILE	INPUT PSD LEVEL, g^2/Hz	FREQUENCY, Hz	DURATION, MINUTES
A	0.005	20 TO 2000	CONTINUOUS
B	0.015	20 TO 2000	3
C	0.05	20 TO 100	1

MTBF VIBRATION APPLICATION

T ± 5 MINUTES	143	170	197	224	251	278	305	332	359	386	413	440
PROFILE	C	B	C	C	B	B	B	C	B	C	B	C

FIGURE 2. Reliability and life testing environmental profile. Continued

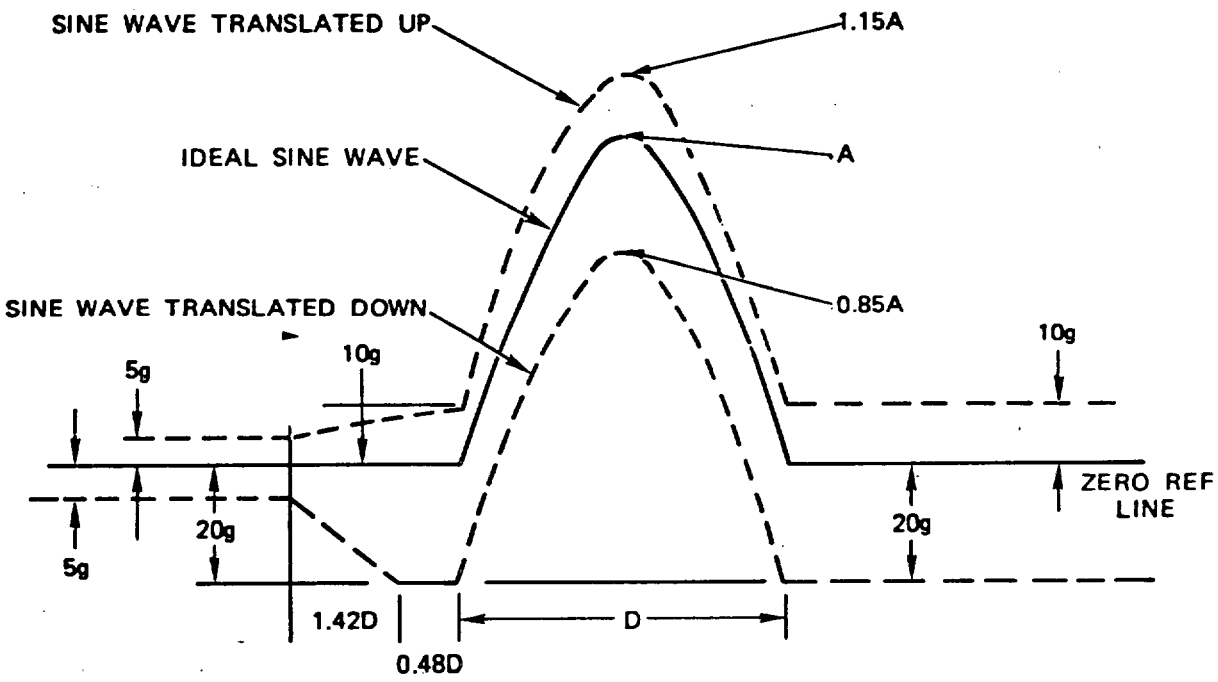
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Type of Work	Cleanliness Note 1	Lighting Foot Candles Note 2	Air Tempera-ture Note 3	Relative Humidity Note 4	Dust Control Note 5	Ventila-tion or Exhaust Note 6	Noise Note 7 dbA	Habitat
Machine Shop								
Tolerance to 0.01	Class D	70	60–100°F	U	Class D	Note 6	90 dbA	Note 8
Tolerance to 0.001	Class C	100	65–95°F	U	Class C	Note 6	80 dbA	Note 8
Tolerance to 0.0001	Class B	200	65–85°F	70% Max.	Class C	Note 6	80 dbA	Note 8
Tolerance finer than 0.0001	Class BA	500	67–75°F	60% Max.	Class B	Note 6	75 dbA	Note 9
Propellant Chemical and Potting Areas	Class B	100	65–95°F	60% Max.	Class B	Note 6	80 dbA	Note 9
Plating and Heat Treating	Class C	70	55–105°F	U	Class C	Note 6	90 dbA	Note 8
Electronic Assembly	Class B	100	65–95°F	30–70%	Class B	Note 6	75 dbA	Note 9
Mechanical Assembly								
Tolerance to 0.05	Class C	100	65–95°F	U	Class C	Note 6	75 dbA	Note 8
Tolerance to 0.001	Class B	100	65–95°F	U	Class C	Note 6	75 dbA	Note 8
Tolerance to 0.0001	Class BA	200	65–85°F	70% Max.	Class B	Not Req	70 dbA	Note 9
Tolerance finer than 0.0001	Class A	500	65–75°F	30–50%	Class A	Not Req	70 dbA	Note 9
Inspection Stations	Equal to product inspected							

1. Cleanliness Definitions
   a. Class D – – Daily Cleanup: Removal of scrap, clean up all spilled oil, etc.
   b. Class C – – Prompt Cleanup: Scrap, oil and residue shall not be allowed to accumulate.
   c. Class B – – Prompt Cleanup: oil, residue, spilled chemicals removed immediately. Floors, walls and work area shall have easily cleaned surfaces. Cleanup of equipment and area shall be accomplished after each 24 hours of operation.
   d. Class BA – – Oil, residue, spilled chemicals, and any foreign material which might develop in the area shall be cleaned immediately. Floors, walls, furniture and work areas shall have hard, grease-resistant, easily cleaned surfaces. Materials to be worked on shall be cleaned of all foreign material prior to acceptance into this area. Flushing or cutting fluids shall be filtered to remove contamination above 30.0 microns. Chips, particles and dust generated during any and all mechanical operation shall be removed during the operation by vacuum. Clean lint-free smocks are required for operators and inspectors. Complete cleaning and wipe down of equipment, tools, fixtures and the area shall be accomplished at the end of each 8 hours of operation or each 24-hour period.
   e. Class A – – Cleanliness controlled in accordance with FED-STD-209, Class 100,000.
2. Lighting
   Indicated values are minimum light intensity values at the work station. Supplemental lighting shall be used when necessary to improve precision and minimize operator fatigue, but brightness ratios within the operators' field-of-view shall not exceed 10 to 1.
3. Air Temperature
   Designated temperature limits are the daily average temperature measurements taken in proximity of the work stations. Ambient temperature shall be between 65°F and 85°F.
4. Relative Humidity
   Designated relative humidity shall be as measured at room ambient temperature. "U" indicates uncontrolled relative humidity.
5. Dust Control Definitions
   a. Class D – – No dust control required.
   b. Class C – – Outside air shall be filtered to remove dust particles. Type of filter is unspecified.
   c. Class B – – Outside and recirculated air shall be filtered to remove dust particles. Filter rating shall be 10 micron maximum.
   d. Class A – – Dust control shall be in accordance with FED-STD-209, Class 100,000.
6. Ventilation or Exhaust
   Forced ventilation or exhaust shall be provided whenever required to minimize operator fatigue.
7. Noise
   Noise is defined as the average sound level existing at the work station during normal operation when measured with a standard sound-level meter as specified in ANSI S1.4-1971 "A" weighted. Work stations for control of automatic equipment operations, which requires operator loading and unloading only, may exceed the noise limits specified by 10 dbA but not to exceed 90 dbA. Inspection and test work stations shall not exceed the values specified. Work areas for mechanical assembly of tolerances finer than 0.001 which utilize laminar flow benches may exceed by 5 dbA the value specified.
8. Habitat
   No eating, drinking or personal grooming is allowed in these work areas. Drinking fountains are permissible.
9. Habitat
   No eating, drinking, smoking or personal grooming is allowed in these work areas. Drinking fountains are permissible.

FIGURE 3. Minimum standards for working environments.

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NOTE:

The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries.

FIGURE 4. Shock test.

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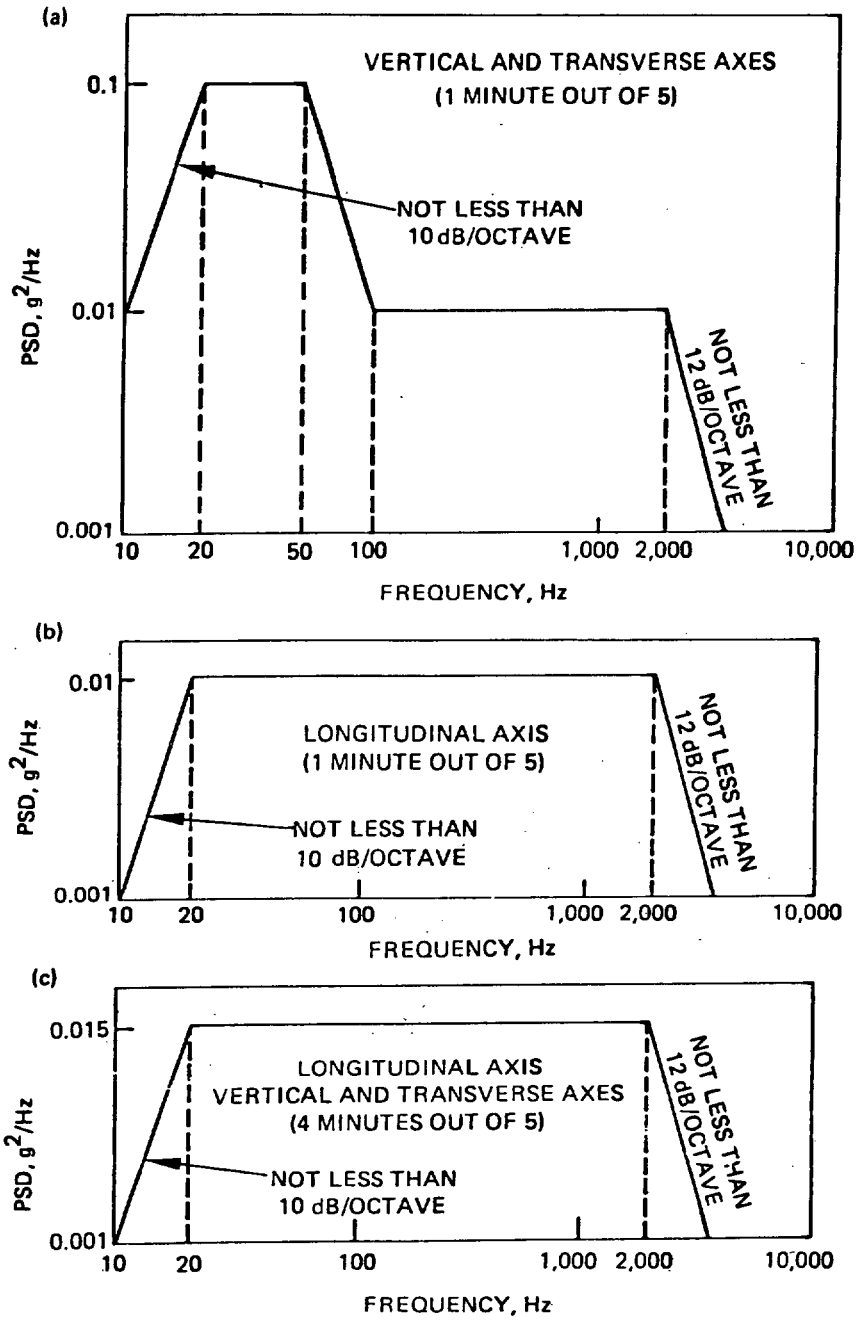
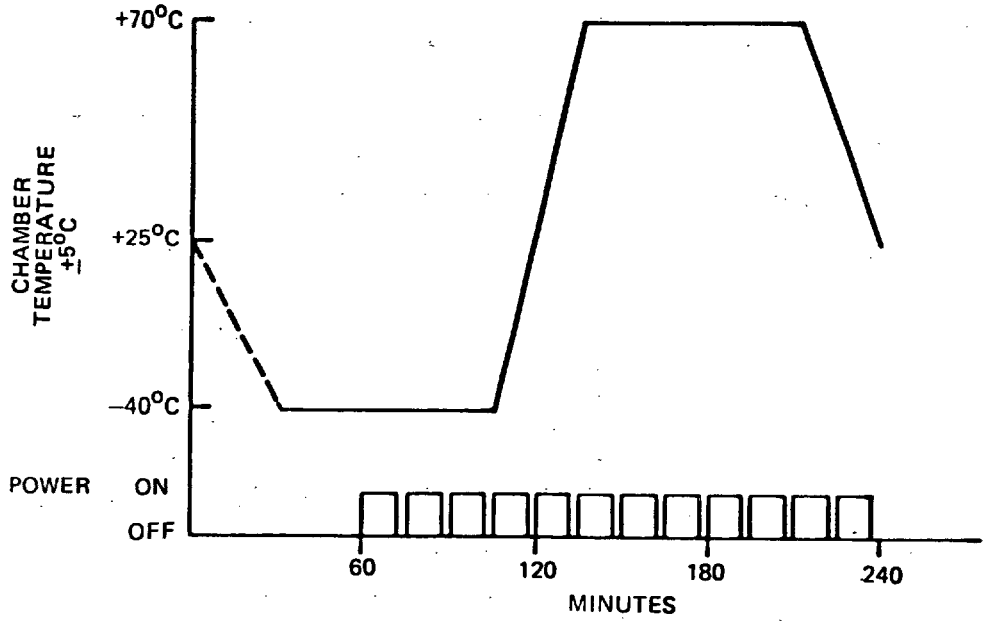


FIGURE 5. Random vibration screen.

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NOTES:

1. POWER IN ACCORDANCE WITH 3.2.1.4.1 SHALL BE CYCLED ON AND OFF BEGINNING AT TIME 60 AS FOLLOWS:
   a. 10 MINUTES ON
   b. 5 MINUTES OFF
2. THE COOLER SHALL MEET THE REQUIREMENTS OF 3.2.1.1 AND 3.2.1.2.2 DURING OPERATING INTERVAL.
3. EACH COOLER SHALL BE SUBJECTED TWO 240 MINUTE CYCLES.

FIGURE 6. Burn-in profile.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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MIL-C-85300 (AS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

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DATE

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